

218_SRIO_Manual

1 Norms and standards

SRIO-11353, ✓ Approved - FCC Compliance

SRIO shall be compliant to FCC Class A.

Remark: "Class A: **Equipment that is**, by virtue of its characteristics, **highly unlikely to be used in a residential environment, including a home business**. Characteristics considered in this assessment include price, marketing and advertising methodology, the degree to which the functional design inhibits applications suitable to residential environments, or any combination of features that would effectively preclude the use of such equipment in a residential environment. Also used for denoting the corresponding emission limits applicable to such equipment."

see also [FCC_Info](#)

SRIO-11351, ✓ Approved - FCC: Compliance Information

In addition to labeling, a compliance information statement shall be supplied with the product.

- Identification of the product as in § 2.1074, e.g., name and model number;
- A compliance statement as applicable, e.g., for devices subject to [part 15 of this chapter](#) as specified in § 15.19(a)(3), that the product complies with the rules; and
- The identification, by name, address and telephone number or Internet contact information, of the responsible party, as defined in [§ 2.909](#). The responsible party for Supplier's Declaration of Conformity must be located within the United States.

see also [FCC_Info](#), example:

APPENDIX A

EXAMPLE OF SUPPLIER'S DECLARATION OF CONFORMITY COMPLIANCE INFORMATION STATEMENT INCLUDED WITH AN END PRODUCT AT THE TIME OF MARKETING OR IMPORTATION²²

NOTE: The Commission does not have a required SDoC format. This is an example only and is provided to illustrate the type of information that may be supplied with the product at the time of marketing or importation for meeting the FCC SDoC requirement Section 2.1077 compliance information.

Supplier's Declaration of Conformity 47 CFR § 2.1077 Compliance Information Unique Identifier: (e.g., Trade Name, Model Number) Responsible Party – U.S. Contact Information ABC Corporation Street Address City, State Zip Code Telephone number or internet contact information FCC Compliance Statement (e.g., products subject to Part 15) This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.
--

SRIO-11348, ✓ Approved - FCC: Statement in Manual

The following text shall be added to manual to inform the user about FCC conformity:

"This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

(1) This device may not cause harmful interference, and

(2) This device must accept any Interference received, including interference that may cause undesired operation."

SRIO-11349, ✓ Approved - FCC: Device Labeling

The labeling of SRIO shall have the FCC Logo as shown below as well as the Canadian certification information added.

Logo	Canadian Certification Information
	CAN ICES (A) NMB (A)

SRIO-11347, ✓ Approved - FCC Remark

The FCC statement  **SRIO-11348 - FCC: Statement in Manual** shall be available for the user inside of the packaging (e.g. written in manual).

Remark:

If an instruction manual is not provided or is only available electronically, then the labeling (Section 15.19(a) statement) shall also be placed on the device packaging, paper insert or on a removable label attached to the device.

SRIO-9317, ✓ Approved - RoHS

SRIO shall be compliant with **Restriction of the use of certain hazardous substances (RoHS) Directive 2011/65/EU**.

SRIO-9320, ✓ Approved - Machinery Directive

SRIO shall be compliant with **Machinery (MD) Directive 2006/42/EC**.

SRIO-9319, ✓ Approved - Low Voltage Directive

SRIO shall be compliant with **Low voltage (LVD) Directive 2014/35/EU**.

SRIO-9318, ✓ Approved - Electromagnetic Compatibility

SRIO shall be compliant with **Electromagnetic Compatibility (EMC) Directive 2014/30/EU**.

SRIO-9316, ✓ Approved - Applicable standards - CE

Derived from directives above, SRIO shall be compliant with the standards as listed in "List of applicable standards.xlsx".

Remark: The list is available at ...\\SRIO\\30_QM_Plan_Testplans\\02_Functional_Safety\\03_Functional Safety Engineering\\CE

SRIO-7134, ✓ Approved - EG-Conformity Declaration

For each device, a copy of the eg-conformity declaration for both languages, english and german, shall be added.

2 List of Work Items

2.1 General

 **SRIO-17099 - Content for User Documentation (derived from Risk Assessment)**

 **SRIO-12352 - Properties of the device which should be defined according to ZVEI CB24I, but are not.**

SRIO-15403, ✓ Approved - Documentation: Info regarding mechanical defects

It shall be documented in Technical Publication that devices with mechanical defects being able to be identified by visual inspection shall be replaced.

SRIO-14714, ✓ Approved - Technical Publication - mechanical defects - devices to be replaced

It shall be documented in Technical Publication that devices with mechanical defects being able to be identified by visual inspection shall be replaced.

SRIO-14717, ✓ Approved - Technical Publication - FE - max Current

It shall be documented in Technical Publication that FE is not intended to carry current, it is used for EMC only.

Remark: Be aware related to the differences between "FE" and "PE".

SRIO-16242, ✓ Approved - Release Notes

For each FW which is released, release notes stating the user relevant changes compared to the previous version shall exist.

SRIO-12749, ✓ Approved - Cable Installation

The cable installation for SRIO for input and output ports shall be indoor installation only.

SRIO-9841, ✓ Approved - Restriction Inhouse

The usage of the device and all belonging cable installation is restricted to indoor installation only.

SRIO-11810, ✓ Approved - Assembly

The device shall be mounted on a flat surface.

SRIO-11809, ✓ Approved - Torsional forces or mechanical loads

Torsional forces or mechanical loads during or after installation by the user are excluded and need not be assumed.

Remark: Add note to manual.

SRIO-11808, ✓ Approved - Technical specifications for power dependence on ambient temperature

The Hardware shall provide derating specifications for power usage on Inputs or Outputs during high environmental temperatures.

SRIO-9691, ✓ Approved - Information about vibration and temperature

The manual shall contain the information on the vibration and temperature profiles.

SRIO-11524, ✓ Approved - SCPU IoT: Doku Current US_DI without

It shall be documented, that the value for current US_DI is only the current used to supply sensors without internal usage.

SRIO-10730, ✓ Approved - Manual remark Reset Sensor

During a reset of a sensor device, the danger area shall be secured.

SRIO-9842, ✓ Approved - Cable Length for IO

The usage of the device is restricted to cable length for input and output ports to less or equal 30m.

SRIO-1085, ✓ Approved - Applicable Cable Length

The applicable cable length for the outputs must be ≤ 30 m

SRIO-9375, ✓ Approved - ETH Module 6x2 F-DI and 2x2 F-DO

The device shall provide an IO-Port Configuration of 6 DI-Ports and 2 DO-Ports.

SRIO-8802, ✓ Approved - LED informational text

A note shall be added to the manual, which restricts the usage of the LED as none-safety related.

SRIO-8120, ✓ Approved - PROFIsafe Safety Manual Requirements

The safety manual shall include the applicable instructions, information and parameters of table 38 of the PROFIsafe specification[7].

SRIO-6557, ✓ Approved - Downgrading Documentation

For Downgrading there shall be a chapter in the safety manual. The respective requirement can be found in the Safety Concept (see  [SRIO-3598 - SC FWU FW Update Process in Safety Manual](#)).

SRIO-15096, ✓ Approved - FW-Update "Downgrading"

In the manual the user shall be informed about the his responsibilities by downgrading the preinstalled FW-Version.

Note: Usecase: Machine of the User is certified with a lower FW-Version and his application does not require him to use the current released FW-Version of SRIO (which is installed in production).

SRIO-7322, ✓ Approved - Packaging

The device shall be packaged in a cardboard box.

SRIO-7321, ✓ Approved - Manual

The device shall be packaged along with a printed manual.

SRIO-6533, ✓ Approved - Installation Description

The reached safety relevant values of EN ISO 13849 as well as IEC 61508 shall be made available to the customer, so that the customer is able to use the device in its own safety application.

SRIO-7135, ✓ Approved - Instructions For Wiring

The user manual shall state the reached SIL Level in respective of the configuration (port usage).

SRIO-1124, ✓ Approved - Timings - Maximum Response Time

There shall be a formular in the user manual how to calculate the maximum response time.

Remark:

Might as well inform about the calculation sheet from Siemens.

SRIO-1976, ✓ Approved - Timings - Maximum Response Time for IOs

In the manual the parameter affecting the calculation of the maximum response time shall be stated.

SRIO-14971, ✓ Approved - DO - Achievable SIL/PL depending on Parametrization of DO

The manual shall state the achievable SIL / PL for the different configuration possibilities of DO.

SRIO-1433, ✗ Rejected - SC PU Time

The safety manual shall state that:

It is within the responsibility of the user that the device does not hold the application in a safe state for longer than xxh outside of its cyclic operation, e.g. leaving the device in power-up, FW-update, configuration or failure state

SRIO-14946, ✓ Approved - PS - Max Current UA

The manual shall a maximum of 16A for the supply UA.

Remark:

This results in a maximum current to daisy chain UA [$I_{UA_DaisyChain_Max}$] of 12A.

*Result from SRIO-8311 (16A per pin Power In) and SRIO-9511 max 4A for 2*DO.*

SRIO-14945, ✓ Approved - PS - Max Current US

The manual shall state a maximum of 16A for the supply US.

Remark:

This results in a maximum current for daisy chain US [$I_{US_DaisyChain_Max}$] of 11.5A

Result from SRIO-8311 (16A per pin Power In) and SRIO-9512 max 3A9 for max current US_DI + SRIO-14947 ~300mA internal usage. [rounded to next .5].

SRIO-8311, ✓ Approved - PS - Max Current Power In Pin

The Manual shall define a maximum of 16A for each Pin of the Power In Plug.

Note: This Current is not measured, since the current drawn via daisy-chaining is unknown to SRIO.

 [SRIO-17003 - SC COM PROFI-safe Safety Manual](#)

 [SRIO-16980 - SC FWU Info in Safety Manual regarding validation of Version](#)

SRIO-8879, ✓ Approved - Actuator - No external connection

The device manual shall contain a note stating that actuators must only be electrically connected via the SRIO interface, the actuator shall not be supplied with power and ground by any other interface.

SRIO-7889, ✓ Approved - Remark External Voltages

The manual shall prohibit the connection of external voltages.

 [SRIO-13938 - LED - Technical Publication - Documentation of LED behavior](#)

 [SRIO-13121 - LED - Technical Publication - LED for information only](#)

 [SRIO-10729 - Commands by CPU3 to SCPU - reactivation of Sensor Type C - Technical Publication](#)

 [SRIO-1280 - Yellow Marked Safety Ports](#)

SRIO-7303, ✗ Rejected - Clearance & creepage distance

The safe Remote IO shall comply with the EN 61010-1 regarding clearance & creepage distance.

2.2 Security

SRIO-16243, ✓ Approved - Security Risk Assessment

A risk assessment regarding security shall be performed.

SRIO-9402, ✓ Approved - Security

The usecase of this device is assumed to be located in a safe zone behind a firewall

SRIO-1196, ✓ Approved - FW-Update Behind Firewall

From the point of view of Security the FW-Update shall be done behind a firewall ("shopfloor") by a trustworthy person. This requirement will be implemented via SRIO's manual and the customer who follows the rules defined in the manual.

2.3 SCPU

-  ~~SRIO-14608~~ - Parametrization -- P3 -- F-Parameter -- Technical Documentation -- F_Source_Add ≠ F_Dest_Add
 -  ~~SRIO-12436~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication --Default Setting -- Testpule Supply
 -  ~~SRIO-12435~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication --Default Setting -- Symmetrie
 -  ~~SRIO-12434~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication --Default Setting -- Resolve Symmetry Violation
 -  ~~SRIO-12432~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication -- DI -- Filter Time -- 3ms default
 -  ~~SRIO-12432~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication -- DI -- Filter Time -- 3ms default
 -  ~~SRIO-12407~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication -- DI -- No Effect -- 1oo1 -- Channel enabled
 -  ~~SRIO-12408~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication -- DI -- No Effect -- 1oo1 -- Channel disabled
 -  ~~SRIO-12406~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication -- DI -- Error -- 1oo1 -- Single Channel enabled -- 1oo2 enabled
 -  ~~SRIO-12404~~ - State Fatal Error - EC1 - Technical Publication - Several Returns to after Cold Start
 -  ~~SRIO-13105~~ - Initialization - State "Off" - Transition Conditions T1
 -  ~~SRIO-12432~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication -- DI -- Filter Time -- 3ms default
 -  ~~SRIO-12124~~ - ZVEI - I-Parameter -- Documentation TechPub
 -  ~~SRIO-12043~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication -- DO PL e only if Test Pulses enabled
 -  ~~SRIO-12042~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication -- DI PL independent from Test Pulses
 -  ~~SRIO-11472~~ - State Initialization - I4 - Reaction based on Number Rotary Switches - 000
-

-  ~~SRIO-12437~~ - Parametrization -- P3 -- I-Parameter -- Technical Publication -- Default Setting -- DI
 -  ~~SRIO-11466~~ - State Parametrization - Reaction based on Number Rotary Switches - 001 to 899
 -  ~~SRIO-11463~~ - State Initialization - I4 - Reaction based on Number Rotary Switches - 999
 -  ~~SRIO-9927~~ - FatalError Remark Manual
 -  ~~SRIO-3598~~ - SC FWU FW Update Process in Safety Manual
-

-  ~~SRIO-12078~~ - I-Parameter - Functions - Filter Time (7 - Filter F-DI 1 & 8 - Filter F-DI 2)
-  ~~SRIO-12085~~ - I-Parameter - Functions - Discrepancy Time - Start of Timer (5 - Symmetry Discrepancy Time)
-  ~~SRIO-12084~~ - I-Parameter - Functions - Discrepancy Time - second channel within (5 - Symmetry Discrepancy Time)
-  ~~SRIO-12083~~ - I-Parameter - Functions - Discrepancy Time - second Channel not within (5 - Symmetry Discrepancy Time)
-  ~~SRIO-12081~~ - I-Parameter - Functions - Discrepancy Time - Restart of Timer (5 - Symmetry Discrepancy Time)
-  ~~SRIO-12027~~ - ZVEI - iParameter -- relationship for Actuator ZVEI Type D
-  ~~SRIO-12029~~ - ZVEI - I-Parameter -- relationship for Sensor ZVEI Type C
-  ~~SRIO-12030~~ - ZVEI - I-Parameter -- relationship for Sensor ZVEI Type A
-  ~~SRIO-12028~~ - ZVEI - I-Parameter -- relationship for Actuator ZVEI Type C

SRIO-9023, Approved - CRC-Tool

An externally developed and certified CRC-Tool shall be used to generate the iPar CRC.

Remark 1: The externally developed Tool is qualified as a T2 off-line support Tool.

Remark 2: The T2 off-line support Tool shall be certified by an accreditation body (TÜV Süd).

2.4 Com

SRIO-7684, ✖ Rejected - Acyclic writing access shall be done only in PLC configuration phase.

The configuration records (defined in GSD) are transferred between param-begin and param-end commands from PLC to device.

The command param-end of PLC indicates that parametrization is finished.

With the param-end command the changes have to be applied.

 **SRIO-715 - Fieldbus Connector Pinout**

 **SRIO-713 - Fieldbus Connector**

2.5 DI

SRIO-15138, ✔ Approved - DI - Achievable SIL/PL - unpulsed, one-channel

	Testpulse	Sensor ZVEI Typ A	OSSD Sensor ZVEI TypC	Safety Performance
1oo1	0	1	0	Cat. 1 / PL c SIL 1

SRIO-15140, ✔ Approved - DI - Achievable SIL/PL - pulsed, one-channel

	Testpulse	Sensor ZVEI Typ A	OSSD Sensor ZVEI TypC	Safety Performance
1oo1	0	0	1	Cat. 2 / PL d SIL 2
1oo1	1	1	0	Cat. 2 / PL d SIL 2

SRIO-15139, ✔ Approved - DI - Achievable SIL/PL - two-channel

	Testpulse	Sensor ZVEI Typ A	OSSD Sensor ZVEI TypC	Safety Performance
1oo2	0	1	0	Cat. 3 / PL d SIL 3
1oo2	1	1	0	Cat. 4 / PL e SIL 3
1oo2	0	0	1	Cat. 4 / PL e SIL 3

SRIO-14969, ✔ Approved - DI - Achievable SIL/PL

The manual shall state the configuration and what SIL/PL the user is able to achieve with this configuration.

SRIO-14503, ✔ Approved - DI - ZVEI Support

DI Module shall support ZVEI-Type A and C

SRIO-12645, ✔ Approved - ZVEI Interface-Typ C [DI]

To comply to ZVEI Interface-Typ C, a table including the required interface parameter shall be included to the manual.

Remark: the blue marked information shall be defined during implementation.

Parameter	min.	typ.	max.
Class	Class 1		
Test Pulse Duration t_i	-	-	Maximum Test Pulse Duration t_{i_max}
Test Pulse Interval T	Minimum Test Pulse Interval T_{min}	-	-
Input Resistance R	Minimum Input Resistance R_{L_min}	-	-
Input Capacity C_L	-	-	Maximum Input Capacity C_{L_max}
Inductance L_L (*)	-	-	Maximum Input Inductance L_{L_max}

SRIO-12643, ✔ Approved - ZVEI Interface-Typ A [DI]

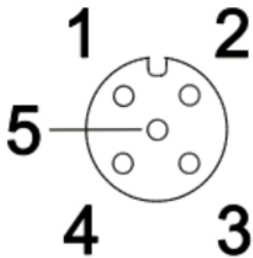
To comply to ZVEI Interface-Typ A, a table including the required interface parameter shall be included to the manual.

Remark: the blue marked information shall be defined during implementation.

Parameter	min.	typ.	max.
Input Current I_i (in ON state)	<i>Minimum Current I_i min</i>	-	<i>Maximum Current I_i max</i>
Output Voltage U_i	<i>Minimum Voltage U_i min</i>	-	<i>Maximum Voltage U_i max</i>
Input Capacity C_i	-	-	<i>Maximum Capacity at Input C_i max</i>

SRIO-12541, Approved - Input Connector Pinout

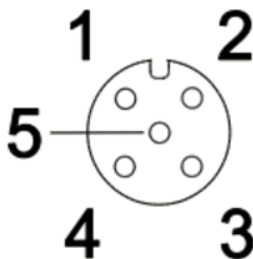
The digital input connector must have the following pinout:



Electrical connection					
Safety Input Port					
Contact Designation	Device Labelling	Function	Function Type	Supply Voltage	Description
1	L+ (US)/ TSOOut1	L+	Supply F-DI1	US	24 V DC (US) Testsignal / Supply related to F-DI1
2	F-DI2	F-DI	F-DI2	US	Safety Digital Input 2
3	L- (US)	L-	GND	US	0 V DC (US)
4	F-DI1	F-DI	F-DI1	US	Safety Digital Input 1
5	L+ (US)/ TSOOut2	L+	Supply F-DI2	US	24 V DC (US) Testsignal / Supply related to F-DI2

SRIO-719, Rejected - Input Connector Pinout

The digital input connector must have the following pinout:



Pin	Signal	Description
1	TSOut1	24 VDC (US) Test/signal out1
2	Safe IN 2	Safety Capable Input 2
3	GND	0 V DC (US)
4	Safe IN 1	Safety Capable Input 1
5	TSOut2	24 VDC (US) Test/signal out2

SRIO-718, Approved - Digital Input Connector

The digital input connector must be of type M12, A-Coded, 5 pins and **female**.

SRIO-11189, ✓ Approved - DI - Inductive Load on Sensor-Supply

The maximum supported inductive load shall be stated in the manual.

SRIO-11193, ✓ Approved - DI - Capacitive Load on Sensor-Supply

The maximum supported capacitive load shall be stated in the manual

Note: Linked Produkt Requirements states a minimum of Cable length 30 m + 20 nF

SRIO-2098, ✓ Approved - Capacitive Load on Sensor-Supply

The supply of the inputs shall support capacitive load of: Cable length 30 m + 20 nF

SRIO-11184, ✓ Approved - Exclusion of external Power at DI-Pins

It shall be documented in Technical Publication (e.g. User Manual) that the DI-Pins (Port/ Channel Pins and Power Supply) shall not be supplied by external voltage.

SRIO-8899, ✓ Approved - No External Power Supply of Sensors

It shall be documented that any external power supply of the Sensors is forbidden.

SRIO-10184, ✓ Approved - Filtersettings Type C

The selected filter time shall be greater than the testpulse of the sensor if ZVEI Type C shall be used.

Remark:

An information shall be added to the manual.

SRIO-8901, ✓ Approved - Type of Sensors

It shall be documented, that Sensors of the ZVEI-Type A or C can be connected.

SRIO-8220, ✗ Rejected - Detection of Over Current for Component DI

Over Current for DI shall be detected in total if $I_{DI} \geq 3A9$.

SRIO-9665, ✓ Approved - DI - Port Error

Errors which are locally restricted for one port, shall not affect other ports.

Remark1: e.g. Short Circuit within the external wiring of a Sensor.

Remark2: Only possible with Error Class 4.

Remark3: Only possible with ChannelGranular Configuration (F_Passivation Parameter).

SRIO-8880, ✓ Approved - DI - SensorSupply

Each DI-Module shall provide two sensor supplies [TSOutx].

SRIO-8035, ✓ Approved - DI - Inputs

Each DI-Module shall provide a safe input port with two input channels.

 ~~SRIO-12862~~ - Test Pulses DI - ZVEI Type A - I-Parameter

 ~~SRIO-12751~~ - Test Pulses DI - ZVEI Type C - I-Parameter

 ~~SRIO-10503~~ - Test Pulses DI - ZVEI Type C - Compliance with ZVEI Category 1 - Documentation in Product Data Sheet

2.6 DO

SRIO-12917, ✓ Approved - ZVEI Interface-Typ D [DO]

To comply to ZVEI Interface-Typ D, a table including the required interface parameter shall be included to the manual.

Remark: the blue marked information shall be defined during implementation.

Parameter	min.	typ.	max.
-----------	------	------	------

Class	Class 1		
Test Pulse Duration t_i	Minimum test pulse duration t_{i_min}	-	Maximum Test Pulse Duration t_{i_max}
Test Pulse Interval T	Minimum Test Pulse Interval T_{min}	-	-
Leakage current $I_{Leakage}$ of the output in OFF state			Maximum output current $I_{Leakage}$
Rated current I_N of the output in ON state	Minimum Input Resistance R_{L_min}	-	Rated current I_N
Capacitive load CL	-	-	Maximum Capacity load C_{L_max}
Inductance L_L	-	-	Maximum Inductance load L_{L_max}

SRIO-12918, Approved - ZVEI Interface-Typ C [DO]

To comply to ZVEI Interface-Typ C, a table including the required interface parameter shall be included to the manual.

Remark: the blue marked information shall be defined during implementation.

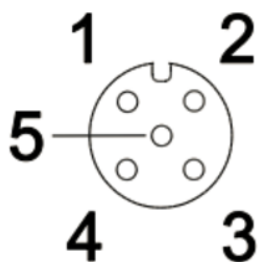
Parameter	min.	typ.	max.
Class	Class 1		
Test pulse duration t_i	Minimum test pulse duration t_{i_min} (*)	-	Maximum Test Pulse Duration t_{i_max}
Test Pulse Interval T	Minimum Test Pulse Interval T_{min} (**)	-	-
Rated current I_N	-	-	Rated current I_N
Capacitive load CL	-	-	Maximum Capacity load C_{L_max}
Inductance L_L (*)	-	-	Maximum Inductance load L_{L_max}

(*) optional

(**) to be stated if t/T is greater than 1%

SRIO-11008, Approved - Digital Output Connector Pinout

The digital output connector shall have the following pinout:



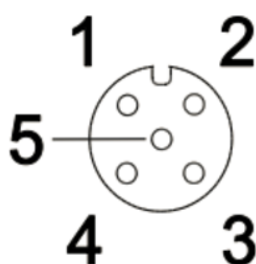
Electrical connection					
Safety Output Port PP-Switching					
Contact Designation	Device Labelling	Function	Function Type	Supply Voltage	Description
1	n.c.	n.c.	-	-	-
2	F-DO2	F-DO	F-DO2	UA	Safety Digital Output 2
3	L- (UA)	L-	GND	UA	0 V DC (UA)
4	F-DO1	F-DO	F-DO1	UA	Safety Digital Output 1
5	FE	FE	FE	FE	Connection to Functional Earth

Electrical connection

Safety Output Port PM-Switching					
Contact Designation	Device Labelling	Function	Function Type	Supply Voltage	Description
1	n.c.	n.c.	-	-	-
2	n.c.	n.c.	-	-	-
3	L- (UA)	L-	GND	UA	0 V DC (UA)
4	F-DO1	F-DO	F-DO1	UA	Safety Digital Output 1
5	FE	FE	FE	FE	Connection to Functional Earth

SRIO-717, ✖ Rejected - Digital Output Connector Pinout

The digital output connector must have the following pinout:



Pin	PP-Switching	PM-Switching
1	NC	NC
2	F-DO 2 (Px)	NC
3	GND	Outx-(Mx)
4	F-DO 1(Px)	Outx-(Px)
5	FE	FE

SRIO-716, ✔ Approved - Digital Output Connector

The digital output connector must be of type M12, A-Coded, 5 pins and **female**.

SRIO-9000, ✔ Approved - DO - Load Inductive - Documentation

The supported inductive load for each output channel shall be declared in the safety manual.

SRIO-9009, ✔ Approved - DO - Load Capacitive - Documentation

The supported capacitive load for each output channel shall be declared in the safety manual.

SRIO-724, ✔ Approved - Diagnosis of Switching Components I

The system shall provide a parameter that enables OR disables the switch-off test pulses for each output port.

SRIO-8982, ✔ Approved - Testpulse Deactivation

It shall be possible to deactivate the switch-off testpulse per output port via parameter.

SRIO-8983, ✔ Approved - Testpulse Deactivation Responsibility User

If the testpulse are deactivated, the user has to take care to switch the channel signals within TD2.

SRIO-7496, ✔ Approved - DO - Maximum Switching Frequency

The SRIO Manual shall contain the maximum switching frequency for the respective loads

SRIO-1968, Approved - Diagnosis of Switching Components II

The safety manual must describe, which measures must be taken when the switch-off test is disabled (see  **SRIO-724 - Diagnosis of Switching Components I**).

For example: The outputs must be tested at least every 8 hours by a switching.

 **SRIO-12863 - Test Pulses DO - ZVEI Type D - I-Parameter**

 **SRIO-12864 - Test Pulses DO - ZVEI Type C - I-Parameter**

 **SRIO-3796 - SC DO Test pulse II**

 **SRIO-2613 - SC DO Switching Freq**

2.7 PowerSupply

SRIO-14932, Rejected - External power supply US - TechPublication - 24V0 nominal

It shall be documented, that SRIO shall be supplied with nominal 24VDC according to EN 61131-2:2017.

SRIO-14931, Rejected - External power supply US - Technical Publication - External current - US

It shall be documented in Technical Publication that for all functions of SRIO related to US, it shall be supplied with up to 16A0 DC.

SRIO-11020, Approved - Connector Pinout

The power connectors must have the following pinout:

Electrical connection					
Power Supply Port					
Contact Designation	Device Labelling	Function	Function Type	Supply Voltage	Description
1	L+ (US)	L+	Supply	US	24 V DC (US)
2	L- (UA)	L-	GND	UA	0 V DC (UA)
3	L- (US)	L-	GND	US	0 V DC (US)
4	L+ (UA)	L+	Supply	UA	24 V DC (UA)
5	FE	FE	FE	FE	Connection to Functional Earth

SRIO-9356, Approved - PS - Isolation Documentation remark

The manual shall have an remark that the SRIO fulfills the required isolation voltage by 61010.

SRIO-9355, Approved - PS - Cable

The manual shall provide information to the user which powercable are compatible to SRIO.

Rationale: Test with different cable showed significant different temperature developement for high current scenarios.

SRIO-8882, Approved - External power supply UA - TechPublication - 24V0 nominal

It shall be documented, that SRIO shall be supplied with nominal 24VDC according to EN 61131-2:2017.

SRIO-8305, Rejected - External power supply UA - Technical Publication - External current - UA

It shall be documented in Technical Publication that for all functions of SRIO related to UA, it shall be supplied with up to 16A0 DC.

Rationale: Limitation to 2A2 per Port -> 2*2A2, Daisy Chain

SRIO-8306, Rejected - External power supply UA - External voltage UA

SRIO shall be supplied with 24V DC, -20%/ +25% resulting in 19V2 to 30V0.

SRIO-8677, Rejected - External power supply US - External voltage US

SRIO shall be supplied with 24V DC, -20%/ +25% resulting in 19V2 to 30V0.

SRIO-8624, ✓ Approved - External Power Supply of SRIO

SRIO shall be able to be powered external by PELV.

SRIO-8622, ✓ Approved - Protection Class

SRIO shall cover the requirements according to protection class III, PELV, DIN EN 61140 VDE0140-1:2016-11.

SRIO-8582, ✗ Rejected - External power supply UA - PELV

The external electrical power supply for UA shall fulfill the requirements according to a PELV output voltage.

SRIO-8730, ✓ Approved - Common FE Power - Pin - Power In and Power Out

A common Functional Earth (FE) shall be provided from Power In to Power Out by pin for power supply.

 SRIO-9003 - No External Fly Back Diode

3 PROFISAFE

https://svn.ifm.com/svn/Safe_Remote_IO/99_Information/Fieldbus/PROFIsafe/PROFIsafe-Environment_2232_V26_Dec15.pdf

6.2 User manual

The EUT shall come with a user manual that allows for proper installation, configuration, parameterization, programming, commissioning, troubleshooting, maintenance, and decommissioning. It shall consider and cover all of the appropriate issues listed in:

- IEC 61131-2, general information to be provided by the manufacturer;
- IEC 61508-2:2010, Annex D (safety manual for compliant items);
- IEC 61508-3:2010, 7.4.2.12 and Annex D (safety manual for compliant items, additional requirements for software elements);
- IEC 61784-3-3, 9.7 (safety manual or user manual for PROFIsafe F-Devices/F-Slaves).

These documents require, as far as applicable, the following items:

- Name and address of the manufacturer (brand, picture mark)
- Type designation or serial number
- Intended use
- General description of the functional safety communication system
- Instructions on planning, installation, and mounting of the safety communication system including drawings, circuit diagrams, and mounting elements
- Instructions for commissioning and operation of the safety communication system
- Instructions for preventive maintenance
- Statements on residual risks
- Instructions on the usage of non-ionizing radiation
- Nominal operating voltage(s) with indication of voltage type and frequency
- Power/current consumption
- SIL claim according IEC 61508. In case of factory automation additionally PL/Category according to ISO 13849-1
- Statements on parameterization, configuration and programming as far as required
- Advice on how to determine the safety function response times required short circuit and overvoltage protection means, as far as applicable
- Operating temperature range
- Ingress protection class (IPxy); if required, separate statements on the individual components
- Rated insulation voltages and the degree of pollution
- Required wiring and functional description of wiring blocks and connectors
- Required safety instructions
- Instructions on how to act in case of faults
- Drawings, circuit diagrams, descriptions, and illustrations required for operation, maintenance, repair, and for the check of correct functioning of the safety bus system
- Warnings with respect to typically possible missuses of the safety communication system

– Proof tests and proof test interval for the safety device